$a_{11} = \langle f_1, f_1 \rangle = \int_{-1}^{1} dt = 2$

$a_{12} = \langle f_1, f_2 \rangle = \int_{-1}^{1} t\, dt = 0 = a_{21}$

$a_{13} = \langle f_1, f_3 \rangle = \int_{-1}^{1} t^2 dt = \frac{2}{3} = a_{31}$

$a_{22} = \langle f_2, f_2 \rangle = \int_{-1}^{1} t^2 dt = \frac{2}{3}$

$a_{23} = \langle f_2, f_3 \rangle = \int_{-1}^{1} t^3 dt = 0 = a_{32}$

$a_{33} = \langle f_3, f_3 \rangle = \int_{-1}^{1} t^4 dt = \frac{2}{5}$

$\therefore \begin{pmatrix} 2 & 0 & 2/3 \\ 0 & 2/3 & 0 \\ 2/3 & 0 & 2/5 \end{pmatrix}$

↳ Let V be a vectorspace over $\mathbb{C}$. Suppose to each pair of vectors $u, v \in V$, there is assigned a complex number denoted by $\langle u, v \rangle$. This function is called a complex inner product on V if it satisfies the following axioms.

I1 $\langle \alpha u_1 + \beta u_2, v \rangle = \alpha \langle u_1, v \rangle + \beta \langle u_2, v \rangle$

I2 $\langle u, v \rangle = \overline{\langle v, u \rangle}$

I3 $\langle u, v \rangle \geq 0$ and $\langle u, u \rangle = 0$ iff $u = 0$.

I4 $\langle u, \alpha v \rangle = \overline{\langle \alpha v, u \rangle}$

$= \bar{\alpha} \overline{\langle v, u \rangle} = \bar{\alpha} \langle u, v \rangle$

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